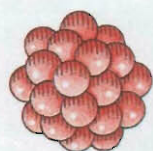


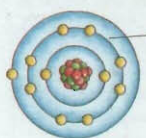
ATOMS AND MOLECULES



TINY PARTICLES CALLED ATOMS are the basic building blocks that make up everything around us. Forces called bonds effectively "cement" the atoms together. A molecule is a cluster of atoms linked by bonds. There are just over a hundred different types of atom, which are themselves made up of even smaller "subatomic" particles, such as protons, neutrons, and electrons.

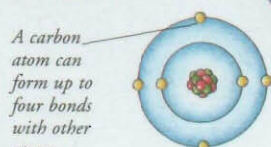
Electron shells and valency

Atoms can have up to seven shells of electrons. An atom with eight electrons in its outermost shell is very stable. Bonds form when atoms gain, lose, or share electrons in order to achieve this stable arrangement. An atom's valency is the number of bonds it can form with other atoms.



Sodium
(3 shells, valency 1)

When sodium bonds, it loses an atom, leaving an outer shell of eight electrons.

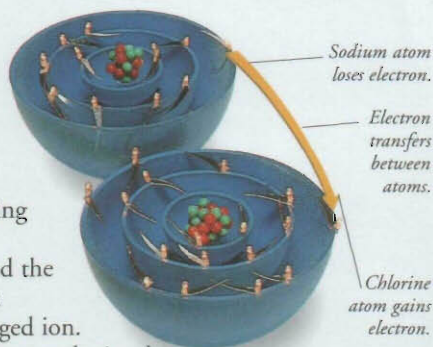


Carbon
(2 shells, valency 4)

A carbon atom can form up to four bonds with other atoms.

Ionic bonds

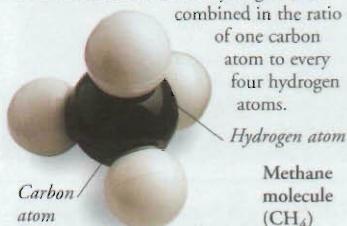
When an electron transfers from one atom to another, the atoms become charged particles called ions. The atom losing the electron becomes a positively charged ion, and the atom gaining the electron becomes a negatively charged ion. The force of attraction between the ions' opposite charges is called an ionic bond.



Formation of ionic bonds in sodium chloride (NaCl)

Chemical formula

Scientists use a kind of code called a chemical formula to describe a substance. The formula uses letters and numbers to show which elements are present in the substance, and in what proportions. Methane, for example, has a chemical formula of CH_4 , which shows that it contains carbon (C) and hydrogen (H), combined in the ratio of one carbon atom to every four hydrogen atoms.



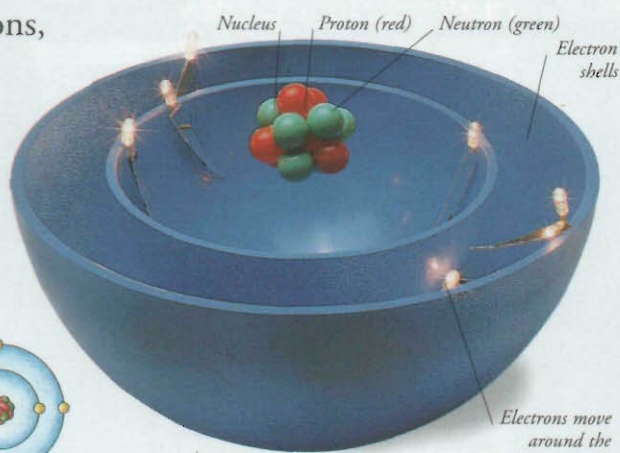
Linus Pauling

The American chemist Linus Pauling (1901–94) won the 1954 Nobel Prize for Chemistry for his work on chemical bonds and the structure of molecules. He calculated the energies needed to make bonds, the angles at which bonds form, and the distances between atoms. He also won the 1962 Nobel Peace Prize for his efforts to stop the testing of nuclear weapons.

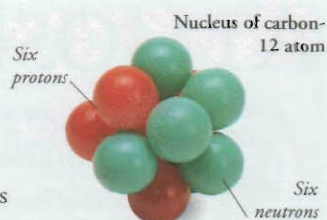


Atomic structure

The centre, or nucleus, of an atom contains particles called protons, which carry a positive electric charge, and neutrons, which carry none. Arranged around the nucleus in layers called shells are negatively charged particles called electrons. The atom has no overall charge, because it contains equal numbers of electrons and protons, so the positive and negative charges are balanced.



Atom of carbon-12 cut in half

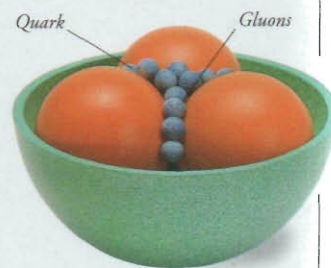


Isotopes

All the atoms of an element have the same number of protons in the nucleus, but some atoms, called isotopes, have different numbers of neutrons. For example, the carbon isotope carbon-12 has six protons and six neutrons, but the isotope carbon-14 has two extra neutrons.

Quarks

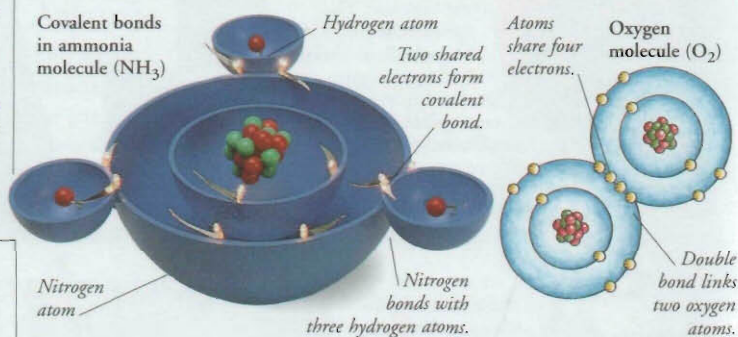
Both neutrons and protons consist of three smaller particles called quarks, stuck together by tiny particles called gluons. Quarks, in turn, may contain even smaller particles.



Inside a neutron

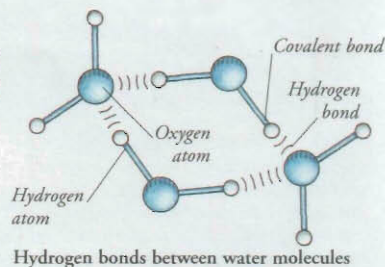
Covalent bonds

A covalent bond forms when two atoms link up by sharing electrons. Each atom supplies an electron, and the pair of electrons orbits the nuclei of both atoms, holding the atoms together as a molecule.



Bonds between molecules

The molecules of covalent compounds are held together by weak bonds called Van der Waal's forces. Some hydrogen-containing compounds, such as water, have stronger forces called hydrogen bonds between their molecules. In water, these bonds form because each oxygen atom in a water molecule is attracted to hydrogen atoms in two nearby molecules.



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ELEMENTS

MIXTURES AND
COMPOUNDS

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POWER

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